ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

INTRODUCTION TO INFORMATION SYSTEMS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Introduction to Information Systems paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept answers in the student's own words if the meaning is correct, and other correct examples.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 I ✓ 1.3.2 F ✓ 1.3.3 C ✓ 1.3.4 H ✓ 1.3.5 E ✓ 1.3.6 D ✓ 1.3.7 B ✓ 1.3.8 J ✓ 1.3.9 G ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Data is raw, unprocessed facts with no meaning on their own ✓, e.g. a list of learners' test marks ✓. Information is data that has been processed and has meaning ✓, e.g. the class average or a ranked mark list ✓. (4)

2.2 Block diagram: Input → Processing → Output ✓✓ (correct order with arrows), with Storage connected to Processing (two-way arrow) ✓✓ (4)

2.3 Any TWO differences (2 marks each): RAM is volatile (loses contents when switched off) ✓ – ROM is non-volatile ✓; RAM can be read and written ✓ – ROM can normally only be read ✓; RAM holds programs and data in use – ROM holds start-up instructions (BIOS/firmware) (4)

2.4 Any THREE with reasons: processor (CPU) speed / number of cores ✓ – faster processing of code and video rendering ✓; amount of RAM (e.g. 16 GB) ✓ – runs several large programs at once without slowing down ✓; storage capacity and type (SSD) ✓ – large video files need space; SSD loads faster ✓; graphics card (GPU) – speeds up video editing; screen size and resolution – detail for editing (6)

2.5 (a) 3 × 1 024 = 3 072 bytes ✓ (b) 2 048 ÷ 1 024 = 2 GB ✓ (2)

[20]

QUESTION 3

3.1 Hardware ✓; software ✓; data ✓; procedures ✓; people ✓ (5)

3.2 Hardware – computers, printers, card readers at reception ✓; software – patient management program / database ✓; data – patient names, ID numbers, medical history ✓; procedures – steps for registering a patient, backing up records ✓; people – clerks, nurses, doctors who use the system ✓ (5)

3.3 A TPS records and processes the day-to-day transactions of an organisation (e.g. sales, bookings) ✓✓; an MIS uses the data from the TPS to produce summary reports that help managers to plan and control ✓✓. (4)

3.4 Any THREE: accurate ✓; relevant ✓; timely / up to date ✓; complete; easy to understand; cost-effective; from a reliable source (3)

3.5 Any THREE: faster to find and process records ✓; saves storage space ✓; easy to back up / fewer records lost ✓; fewer calculation errors; data can be shared by many users; reports produced automatically (3)

[20]

QUESTION 4

4.1 System software controls and manages the computer hardware ✓, e.g. Windows / a device driver ✓. Application software lets the user do a specific task ✓, e.g. Microsoft Word / Excel / a web browser ✓. (4)

4.2 Any THREE: manages memory ✓; manages files and folders ✓; controls input and output devices ✓; provides a user interface; manages processes / multitasking; security (user accounts) (3)

4.3 4.3.1 Assignment1.docx ✓ 4.3.2 C: drive ✓ 4.3.3 Level2 ✓ 4.3.4 Microsoft Word (a word processor) ✓ (4)

4.4 You can find files quickly later ✓ and know what a file contains without opening it / avoid overwriting the wrong file ✓. (2)

4.5 Save stores the changes in the existing file under the same name and location ✓; Save As lets you choose a new name, location or file type, creating a new copy ✓. (2)

[15]

QUESTION 5

5.1 Any TWO: Wi-Fi ✓; mobile data (3G/4G/5G modem or hotspot) ✓; fibre or ADSL with a router; network cable (Ethernet) at college (2)

5.2 Any FOUR: use a clear subject line ✓; start with a proper greeting ✓; keep the message short and to the point ✓; use correct spelling and grammar, no SMS language ✓; do not type in capital letters (shouting); end with your name / signature; check attachments before sending (4)

5.3 Any THREE: install antivirus software and keep it updated ✓; do not open attachments or links from unknown senders ✓; keep the operating system updated ✓; use a firewall; download only from trusted sites; scan flash drives (3)

5.4 Copying someone else's work and presenting it as your own ✓; avoid it by using your own words and acknowledging (referencing) the sources ✓. (2)

5.5 Any TWO with prevention: eye strain ✓ – take regular breaks, adjust brightness / 20-20-20 rule ✓; back and neck pain ✓ – use an adjustable chair, sit upright, screen at eye level ✓; RSI – wrist support, correct keyboard position (4)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE